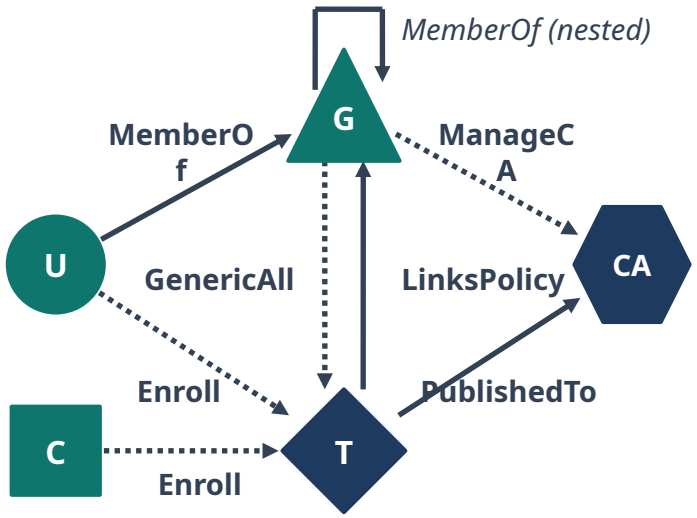
Methodological Workflow: The CertGraph Pipeline

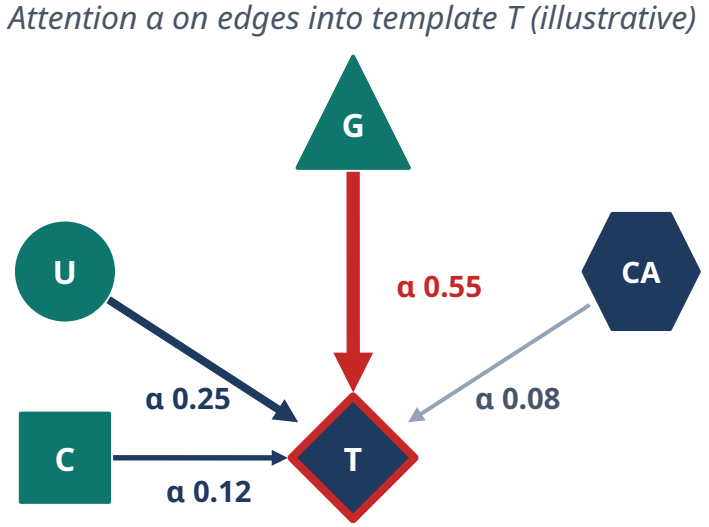
From raw directory ingestion to heterogeneous GNN screening and neuro-symbolic defense

1. Ingest & Build Metagraph



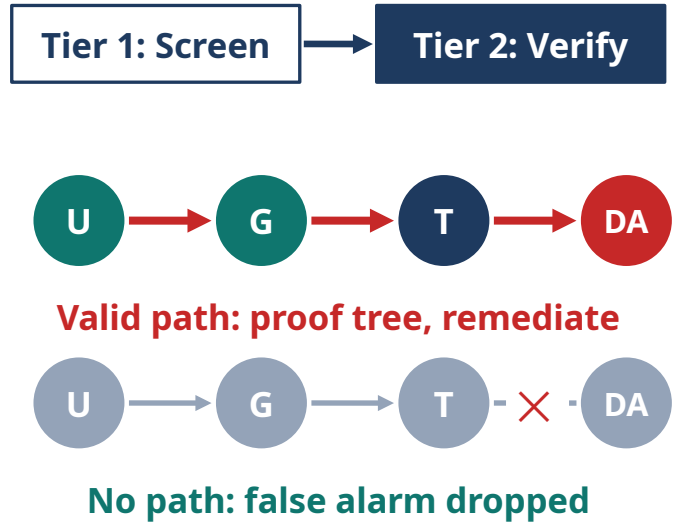
- Ingests LDAP objects and DACLs
- 5 entity classes, 16 relations
- Precomputes group closures

2. Relational GNN Screening



- 64-D type-specific projections
- 2 attention layers, 4 heads
- O(1) risk inference per template

3. Verify & Defend

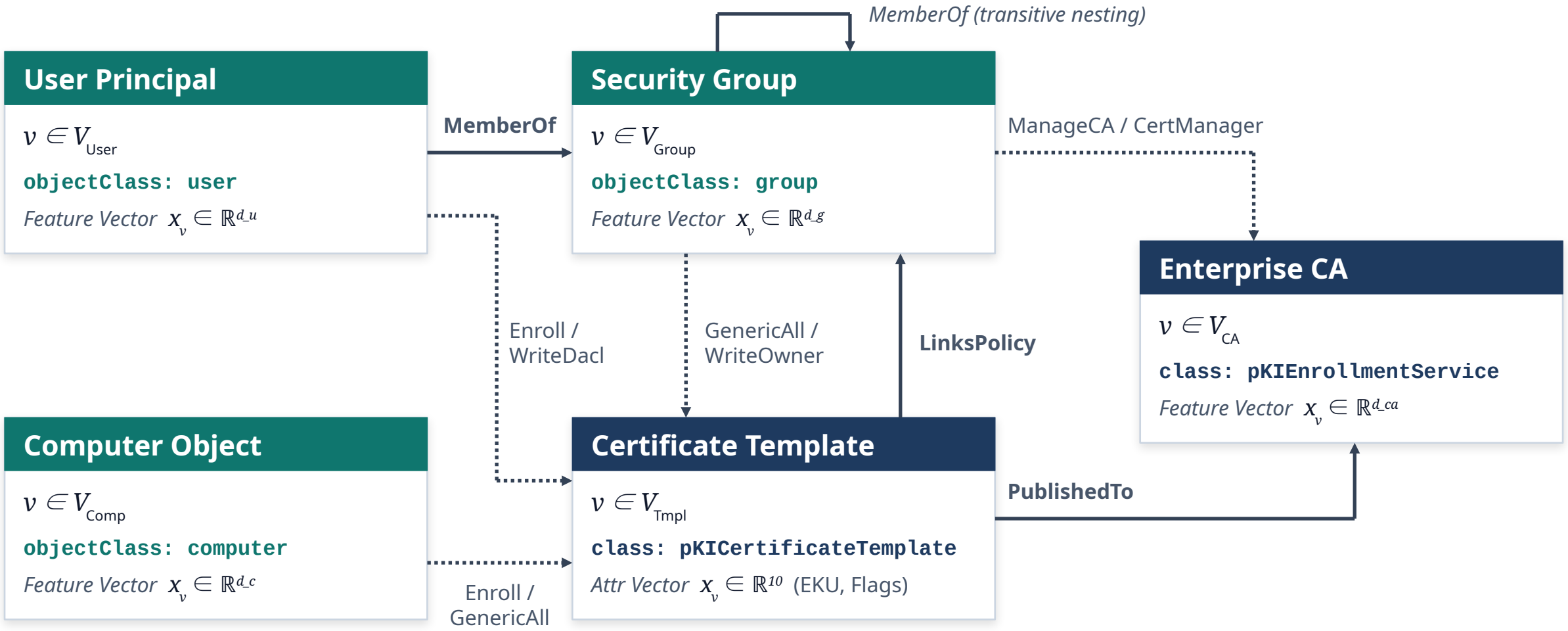


- Symbolic BFS reachability check
- 0 false positives
- Stackelberg edge interdiction

CertGraph unifies neural screening, symbolic verification, and autonomous defense.

Active Directory Metagraph Schema

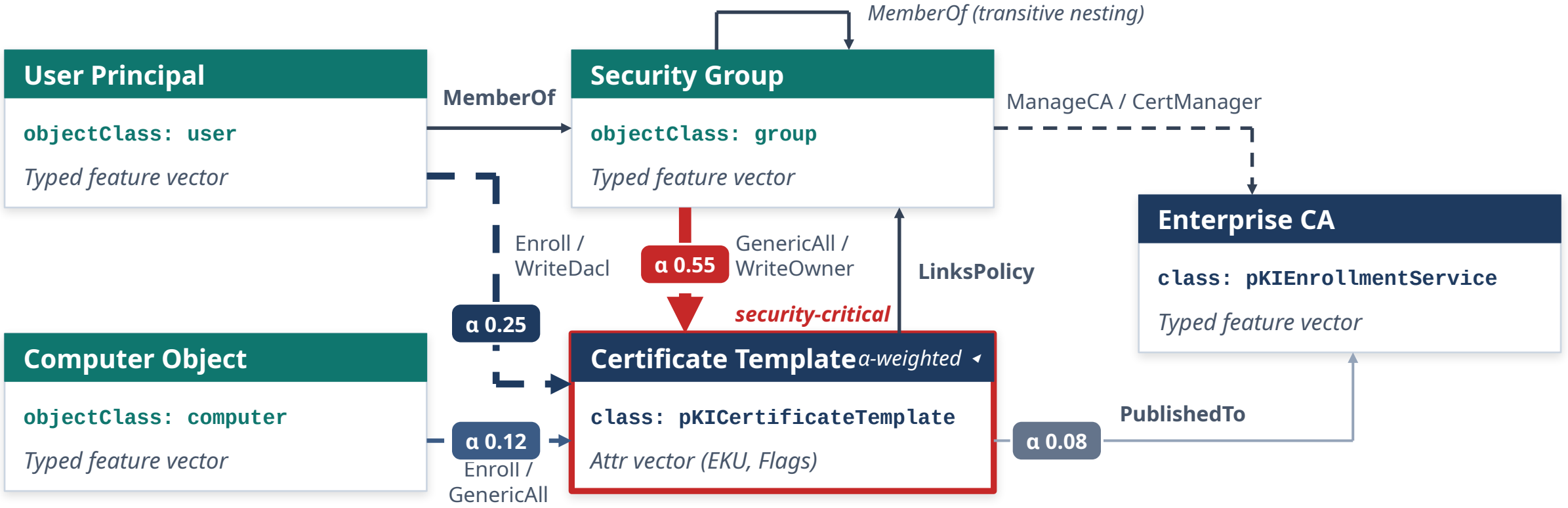
5 entity classes · 16 canonical forward/reverse relations



Active Directory Metagraph: The CertGraph Pipeline

5 entity classes · 16 canonical forward/reverse relations

AD relation · Admin ACL · Thicker = higher attention α



1 Ingest & Build

LDAP + DACLs into 5 classes and 16 relations; group closures precomputed

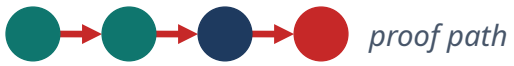
2 GNN Screening: Relational Attention

Softmax over edges into the template · 4 heads · illustrative



3 Verify & Defend

BFS proof (0 FP), then edge cut



The Multi-Hop Attack Challenge: ESC13 Case Study

Diagnostic benchmark: why flat scanners fail on composite privilege escalation paths

External vector: SpecterOps / Oliveau 2024 (Thesis §2.6)



Certipy Heuristic rule scanner

0% detection

Inspects objects in isolation; no single object is misconfigured

BloodHound Deterministic graph search

State explosion

Exhaustive BFS breaks down beyond 100K entities

CertGraph (ours) Relational GNN

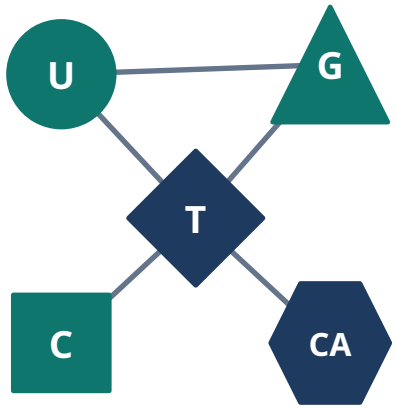
343 ms

Learns multi-hop path closures and detects the full chain

Every permission is benign in isolation. The threat exists only as a multi-hop path closure.

Hetero-GAT Pipeline for ESC Classification

1. Input Graph



5 entity node types

User, Computer, Group, Template, CA

Linear projection

Per node type

2. Hetero-GAT Layer 1

Relational attention

Multi-head (4 heads) over each relation type

Aggregate & normalize

ELU activation + LayerNorm

Residual skip

Avoids source collapse

3. Layer 2 & Readout

2-hop aggregation

Aggregates multi-hop DACLs and policy linkage contexts

Template readout

64-d embedding of each certificate template (isolated vulnerability latent state)

4. Classification

MLP head

64 → 32 → 7 classes

Safe	Benign
ESC1	Enrollee supplies SAN
ESC2	Any Purpose EKU
ESC3	Cert Request Agent
ESC4	DACL / WriteOwner
ESC9	No Security Extension
ESC13	**Policy OID linkage**

End-to-End Detection & Remediation Pipeline

